

Ricardo (Zhicong) Xie

Manufacturing / Site Project Intern | CAD-to-Part, Workshop Practice and Coordination

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LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027 | Open to NZ-wide roles

PROFILE

- Mechanical master's student with laser cutting, CNC exposure, 3D printing workflow and CAD-to-part evidence.
- Interested in manufacturing, production support and site/project engineering roles.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Manufacturing Processes, CNC Machining, Design for Manufacture, Manufacturing and Industrial Systems

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led project planning, task split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Led a small automotive-structure walkthrough covering seat/interior systems and vehicle context.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, validation workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Screened corner-load, strap-lug and screw-boss regions with an FEA-style check.
- Built Keil C firmware structure with sensor, battery and state-machine modules.

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

Team Systems Engineering Feasibility Study

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led systems-engineering structure for a 5-person technical feasibility study.
- Built stakeholder map, requirements hierarchy, ConOps, trade criteria and risk register.
- Used verification gates to keep orbit, power, thermal, communications and cost assumptions aligned.

TECHNICAL SKILLS

- CAD-to-part: AutoCAD/DXF laser-cut prep, SolidWorks assemblies, STEP/STL export and checks.
- Workshop: CNC workflow, workholding, cutting fluid, chip formation and 3D printing workflow.
- Site support: technician communication, task tracking, workshop safety awareness and verification.